

SYNAPSE: SYNTHETIC NETWORK-TRAFFIC AGENT PIPELINE FOR FEDERATED EDGE SECURITY

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ABSTRACT Ensuring federated edge networks' security from intrusion attacks creates a complex problem in itself, wherein distributed nodes need to be provided with vast amounts of diverse traffic, which is stratified by protocols and needs labeling, and yet real traffic is limited due to its legal complications. Existing synthetic traffic generators address this gap inadequately. Statistical generators reproduce marginal feature distributions but discard the causal network behaviors that make attacks detectable. Static-template generators produce causally valid traffic but collapse to narrow regions of the feature space, causing federated models to overfit to fixed tool parameters rather than learning generalizable detection boundaries. Neither class of generator addresses the schema mismatches, non-independent and identically distributed partitioning requirements, or threshold calibration mismatches that arise specifically in federated deployment. This paper introduces SYNAPSE, a Synthetic Network-Traffic Agent Pipeline for federated Edge Security. SYNAPSE coordinates four tightly coupled stages: an agentic synthetic generation engine driven by a large language model operating within a Planner-Configurer-Executor-Reflector state machine over live ContainerNet network emulation; a deterministic schema adapter that bridges the 82-dimensional CIC-IDS2017 feature space to the 66-dimensional protocol-partitioned detector input; a federated detection mesh of four edge nodes aggregated via FedProx with trust-weighted aggregation under Dirichlet-partitioned non-independent and identically distributed data; and a privacy-preserving intelligence layer combining local differential privacy noise injection with an agentic retrieval-augmented generation reporting module. Across volumetric attack scenarios, SYNAPSE achieves a intrusion detection rate of 99.5 percent and a five-class detection F1 of 70.4 percent, while reducing the false positive rate by 18.3 percentage points relative to a static-template baseline. Federated convergence reaches 98.4 percent global accuracy within the target round budget, and the privacy-preserving layer retains 96.8 percent of detection F1 under a formal differential privacy guarantee.

INDEX TERMS federated learning, edge network security, synthetic network traffic, network intrusion detection, network traffic generation, federated edge computing, anomaly detection

I. INTRODUCTION

THE massive proliferation of edge computing devices has radically transformed the cybersecurity landscape. The billions of resource-limited devices that store sensitive information on the edge of the network become potential points of entry for malicious actors. Federated learning has been proposed as a promising architecture for learning IDS models on distributed devices without aggregating the traffic data from multiple devices centrally [1]. In this architecture, the devices keep their locally stored data but send model parameters to a globally running aggregator node. This approach offers the benefits of privacy and distributed learning. However, it also

makes federated architectures vulnerable to attacks which centralized architectures are not susceptible to: non-IID data among devices, Byzantine devices, and the lack of a trusted curator that can check the quality or representativeness of local training corpora [2]. It would require that each node in the proposed federated learning approach would need to learn on traffic that is diverse, causally valid, and truly represents the diversity of attack morphologies possible in practice.

One such challenge needs to be overcome: the consistent lack of annotated and representative network traffic datasets. Various public datasets have contributed greatly to IDS development, especially when dealing with classification tasks,

NSL-KDD, UNSW-NB15, and CIC-IDS2017 [3] being some of them. However, the drawbacks of these datasets are also widely recognized. Traffic contained in them has been collected at specific points in time and limited environments, thus inducing potential biases in any machine learning models based on the aforementioned datasets. Additionally, it is impossible to construct such corpora under novel attack vectors due to ever-changing threats. Also, there are additional obstacles related to privacy concerns: network traffic could contain personal information, rendering reuse of such datasets impossible. As a result, research efforts in federated IDS often end up with models trained and evaluated on very non-representative traffic.

The use of synthetic traffic generation emerges as an obvious solution for this problem. Statistical generators such as CTGAN [4] and variational autoencoders learn marginal distributions of the features from the reference datasets and generate samples from the learned distribution. However, the output traffic is only statistically realistic as it ignores the causality structure, which allows detecting attack traffic: a port scan results in correlated patterns in TCP flags, port numbers, and inter-arrival timings that statistical generators cannot recreate. Generative adversarial networks such as MirageNet [5] and STAN [6] perform better compared to statistical baselines as their output is calibrated against a predefined schema of traffic features but remains inconsistent with the feature representations learned by federated detectors. Schoen et al. [7] prove that generators based on generative adversarial networks fail systematically under distribution shift, while Farah et al. [8] prove that realistic attack traffic can be synthesized only with a clear definition of its causality structure. On the other hand, static-template generators produce causally consistent data points but only within limited regions of the feature space.

Limitations lead to the motivation of a completely new paradigm. Instead of modeling a generative model on a static dataset, an agent-based pipeline allows for control over network simulation tools through the guidance of the LLM, generating causally valid traffic as it travels through the real protocol stack. The agent can diversify attack parameters across jobs, partition the resulting data into non-IID shards that exercise federated optimization mechanisms, and translate the output through a deterministic schema adapter that bridges the generator's native feature space and the detector's input representation. This design directly targets the four compounding problems identified above: causal fidelity, schema alignment, federated suitability, and calibration-deployment mismatch.

This paper introduces SYNAPSE, the Synthetic Network-Traffic Agent Pipeline for Federated Edge Security. SYNAPSE integrates four coordinated stages into a single reproducible pipeline. As shown in Fig. 1, the architecture comprises: (1) an agentic synthetic generation module built on a LangGraph state machine with Planner, Configurer, Executor, and Reflector nodes operating over ContainerNet network emulation; (2) a deterministic schema adapter that

translates 82-dimensional CIC-IDS2017 feature vectors into the 66-dimensional ACTIS representation consumed by the federated detector; (3) a federated detection mesh in which four edge nodes each run per-protocol autoencoders aggregated via FedProx with trust weighting and Dirichlet partitioning; and (4) a privacy-preserving intelligence module that combines a distilled LLM-based PII scrubber with agentic retrieval-augmented generation (RAG) to produce actionable, zero-leak threat reports.

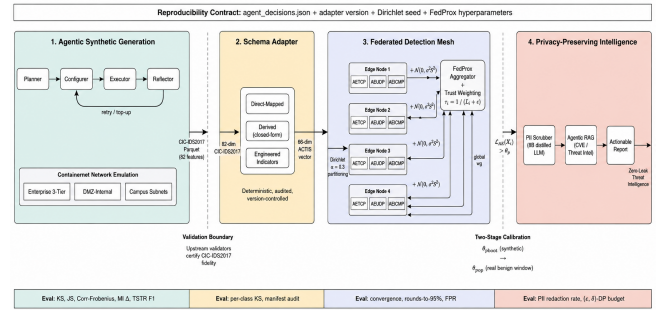


FIGURE 1. Architecture diagram of the proposed four-stage federated intrusion detection system encompassing agentic synthetic generation, schema adaptation, federated detection mesh, and privacy-preserving intelligence modules.

The contributions of this paper are fourfold. First, SYNAPSE introduces an LLM-orchestrated agentic generation loop that produces causally valid, parameter-diverse network traffic through live emulation rather than distributional sampling. Second, the pipeline regards schema translation as an essential methodology component offering versioning, auditing, and a validated adapter. Third, a two-step calibration process rectifies the threshold disparity resulting from applying autoencoder detectors trained on synthesized traffic against the actual traffic. Finally, the privacy-aware intelligence layer provides privacy-respecting threat reports by excluding any form of personal identifiable information (PII) from the telemetry data.

This paper is structured as follows. In the Related Work section, we survey the existing works regarding federated intrusion detection, synthetic traffic generation, and agentic AI security applications. In the Experimental Setup and Methodology section, we describe each stage in detail. In the Evaluation Criteria and Results section, we present the quantitative evaluation results. Finally, the Conclusion and Future Works section contains our conclusion and future work suggestions.

II. RELATED WORKS

A. FEDERATED LEARNING FOR INTRUSION DETECTION SYSTEMS

FL represents an attractive approach for intrusion detection in federated networks, as it allows multiple edge devices to collaboratively train a common model without requiring centralised traffic capture datasets. Survey papers by Khraisat et al. [9] and Wardana and Sukarno [10] classify existing

FL-based IDS solutions along multiple dimensions such as architecture type, aggregation algorithms employed, and open research questions, finding that the major challenge facing FL-IDS is no longer algorithmic in nature, but rather data-related: non-homogeneous, non-independently and identically distributed (non-IID) data negatively impacts model convergence and increases false positive rate. Zhang et al. [11] expand on this topic, demonstrating that FedProx-type of proximal regularization technique partly mitigates client drifts under non-IID setting, although the effectiveness strongly depends on the level of imbalance in labels between nodes. Finally, Dhule and Bansode [12] discuss deep learning approaches implemented in federated pipeline, pinpointing autoencoder-based anomaly detection techniques as effective for detecting novel attacks without labelled examples during training.

More recent applications have also validated these results. In a study by Baidar et al. [13], a hybrid deep learning and FL-based framework was utilized on the IoT and 5G edge network, and it was found that stratifying models based on protocols improved performance compared to using only a single global model especially when local traffic distribution differs. Bilal et al. [14] review constraints on explainability of FL-IDS frameworks within the IoT environment and find that trust-based aggregation methods enhance robustness against poor local updates, although accurate per-client loss estimation is needed. Hakeem and Kim [2] review FL security in vehicle-to-everything (V2X) networks and show how the non-IID problem is further complicated in such networks due to changing topology due to movement. Zhao et al. [15] combine a generative adversarial network (GAN) and FL-IDS to synthetically expand minority classes of attacks prior to aggregation. S. and Manjunath [16] review the architecture of FL-based solutions focused on automotive cybersecurity, with data scarcity and inconsistency in schemas being recognized as the two most commonly mentioned challenges impeding deployment in practice. Melo et al. [17] introduce Anomaly-Flow, an approach to DDoS attacks based on federated GANs working across multiple domains, and prove the effectiveness of federated GANs in heterogeneous environments where no traffic is exchanged. Overall, the aforementioned literature indicates that the progress achieved by FL-IDS research has been primarily methodological; however, access to various protocols still represents a challenge.

B. SYNTHETIC TRAFFIC GENERATION

There have been three major strategies in creating artificial network traffic flows using synthetic network traffic generation techniques. Statistical approaches, which include Conditional Tabular Generative Adversarial Networks (CTGAN) and Variational AutoEncoders (VAE), utilize marginal distribution learning on reference data and generate from learned distribution. Schoen et al. [7] perform an analysis of performance comparison between Generative Adversarial Networks (GAN) and Bayesian Networks in generating network traffic and conclude that while the former successfully

generates the individual statistics of each feature, it cannot preserve any dependencies between those features, such as correlation of some TCP flags combinations and port scans. Nukavarapu et al. [5] introduce MirageNet - a GAN based generator for artificial network traffic generation and state that the output of GAN models resides in low-dimensional manifold and does not capture the diversity of attacks.

The authors present STAN (Sequential Temporal Network), a sequential neural model to generate synthetic network traffic with temporal conditioning, mitigating the shortcomings of inter-arrival time in fidelity somewhat. SyNIG (Synthetic Network Image Generator) is an approach suggested by Sivaroopan et al. [18] where traffic time series data is first mapped to images and then fed into convolutional neural networks to generate data. This results in increased temporal correlation, but decreased interpretability in the data generated. IoTGemini [19] is a traffic generator introduced by Li et al., based on the idea of behaviour-awareness to simulate IoT traffic. IoTGemini learns behavioural models of devices, thus, providing significantly lower distribution discrepancy between the generated and the actual traffic when conditioned upon behavioural characteristics of devices. BehDiffNet [20], introduced by Jiang et al., is a hierarchical diffusion model for behaviour-aware traffic generation. It generates traffic with multi-step behavioural conditioning, resulting in flow-level statistics that exhibit high fidelity.

GAN-based approaches for cybersecurity data augmentation have also been explored in domain-specific contexts. Ammara et al. [21] apply GAN-based generation to SCADA traffic in smart grids, highlighting the challenge of preserving protocol-level causal structure in operational technology environments. Medury et al. [22] propose GANFUSION, which injects GAN-synthesised flows to obfuscate traffic analysis, demonstrating that high-fidelity synthetic flows can deceive ML-based classifiers, an observation that underscores the importance of causal realism in synthetic data used for detector training. In their survey of GAN usage for cybersecurity purposes, Krishnan et al. [23] indicate the dominant problem with GAN-based models in being vulnerable to mode collapse, leading to the creation of fake data sets that only encompass a fraction of the possible attacks. In their survey of techniques for generating synthetic data for privacy-preserving cybersecurity studies, Rapheal et al. [24] suggest that currently there are no generators that satisfy all three criteria (causal fidelity, schema compatibility, and federated). This paper aims to fill this void by basing the generation process on live cyber-network simulation.

C. PRIVACY-PRESERVING ANOMALY DETECTION AT THE EDGE

Edge-based privacy-preserving anomaly detection has been extensively explored due to the increasing adoption of IoT technology. In their paper, Mehnaez and Bertino [25] present an initial model where real-time anomaly detection takes place in an edge device environment without sending data directly to a central server; local training with regular model

synchronization proves to be able to attain a detection rate equivalent to the centralized models. Ma et al. [26] further explore privacy-preserving methods for cloud manufacturing by applying a transformer-based federated learning scheme and find that the use of differential privacy adds a controllable loss of accuracy to it.

More advanced works have been developed towards resource-efficient techniques for such settings. Chen et al. [27] present a lightweight time-series anomaly detector for edge devices in the industrial IoT context, where the algorithm is designed based on model compression techniques, along with providing the privacy guarantee through DP, resulting in good privacy/utility trade-offs at less than a megabyte of model size. Dey et al. [28] investigate the use of neuromorphic computing for implementing an adaptive privacy-preserving anomaly detection technique, discussing energy-efficient improvements pertinent to limited battery-powered edge deployment environments. Asiri et al. [29] develop a FL-based approach for coordinating anomaly detection processes between IoT edge devices using game-theoretic approaches, ensuring the confidentiality of local data privacy in the process. Tang [30] has proposed RedPO-BRNNNet, a federated anomaly detector design using both secure aggregation and differential privacy, arguing that the application of both is needed to avoid inference attacks on the aggregated model. Hendaoui et al. [31] present EPAD for implementing an anomaly detection pipeline based on homomorphic encryption, allowing for secure inference on encrypted edge data. DUDPA-TATD [32] is a lightweight approach to privacy-preserving anomaly traffic detection tailored for edge computing environments, showing that the application of feature-space dimensionality reduction with local DP leads to efficient detection performance and lower communication cost. FedDyGNN-Mesh [33] exploits the concept of federated dynamic GNNs for IoT edge meshes to detect anomalies in their traffic patterns, considering topological dependencies that scalar vectors cannot capture. Together, these studies demonstrate that achieving an optimal privacy utility trade-off is possible at the edge but needs deliberate attention, hence the motivation for the two-step threshold tuning strategy of SYNAPSE.

D. BENCHMARK DATASETS AND THEIR LIMITATIONS

The community for detecting attacks within networks is characterised by a limited number of benchmark datasets, each exhibiting well-documented shortcomings. The NSL-KDD, for instance, was obtained from the famous 1999 DARPA capture and involves artificial traffic simulated in laboratory conditions that does not exhibit the statistical characteristics of current-day networking protocols [34]. Although the UNSW-NB15 offers greater protocol variety than the NSL-KDD dataset, its single organisational source ensures that its class distribution cannot be reliably extrapolated into other types of settings (e.g., enterprise networks and IoT infrastructures) [35]. Meanwhile, the CIC-IDS2017, which utilises CICFlowMeter traffic generation, has become an industry-

standard FL-IDS testbed, despite being riddled with issues related to inconsistent labelling, duplicate flows, and incorrect feature extraction techniques that increase detection accuracy figures artificially [36], [37].

The work of Shafi et al. [38] introduces NTLFlowLyzer and a corresponding large-scale data set in order to solve the problem of inconsistent feature extraction in CICFlowMeter-generated datasets, revealing the inconsistency in flow level features extracted by different tools from the same PCAP, an issue highly relevant to the schema adaptation challenge addressed in the SYNAPSE benchmarking framework. The authors of HyBRID-18 [37] present HyBRID-18, a realistic dataset with rich features designed to fill the realism gap in current benchmarks, yet pointing out that even with well-designed captures, it is impossible to include all attack behaviors a deployed model would have to deal with. Finally, John and Bah [39] review methods for intrusion detection using machine learning and conclude that performance estimates based on public benchmarks greatly overestimate real-world performance due to distributional differences between the two settings. As Hamsar et al. [40] demonstrate through their use of frequent pattern mining on network traffic log datasets, the inherent issue of class imbalance that is seen in all of the significant publicly available benchmark datasets always results in a biased learning boundary towards the predominant benign class. The problems outlined above ultimately highlight the requirement for a well-formulated approach to generating synthetic datasets as needed.

III. EXPERIMENTAL SETUP & METHODOLOGY

A. SYNAPSE PIPELINE ARCHITECTURE

Four inter-dependent steps have been combined together to form a single coherent system known as the SYNAPSE pipeline, which comprises: agent based traffic generation; adaptive schema processing; federated detection mesh functionality and private intelligence reporting. Figure 1 depicts the architecture for the method as a whole, with annotations for reproducibility contracts including the shared run seed, the specific adapter version, and the Dirichlet partitioning seed. All steps provide for auditable output, and the entire pipeline is repeatable solely through the agent decision log.

B. STAGE 1: MULTI-AGENT TRAFFIC SYNTHESIS MODULE

Traffic synthesis stage includes the LangGraph state machine that consists of four consecutive states - Planner, Configurer, Executor, and Reflector. Fig. 2 shows the complete state machine for the agent, which includes conditional loops for retries, along with the four-level validation gate for acceptance.

The Planner takes an initial generation goal specification in the form of $G = (\beta, \pi, S)$, with β being the aggregate budget for flows, π indicating desired category class distributions, and S denoting the target schema destination. It splits the goal G down into a series of attack jobs $\{J_1, \dots, J_n\}$, with a sub-budget β_i such that $\sum_i \beta_i = \beta$ allocated to each job along with a topology chosen from the group $\{T_{\text{enterprise}}, T_{\text{DMZ}}, T_{\text{campus}}\}$.

Fig. 2. LangGraph agent state machine and validation gate

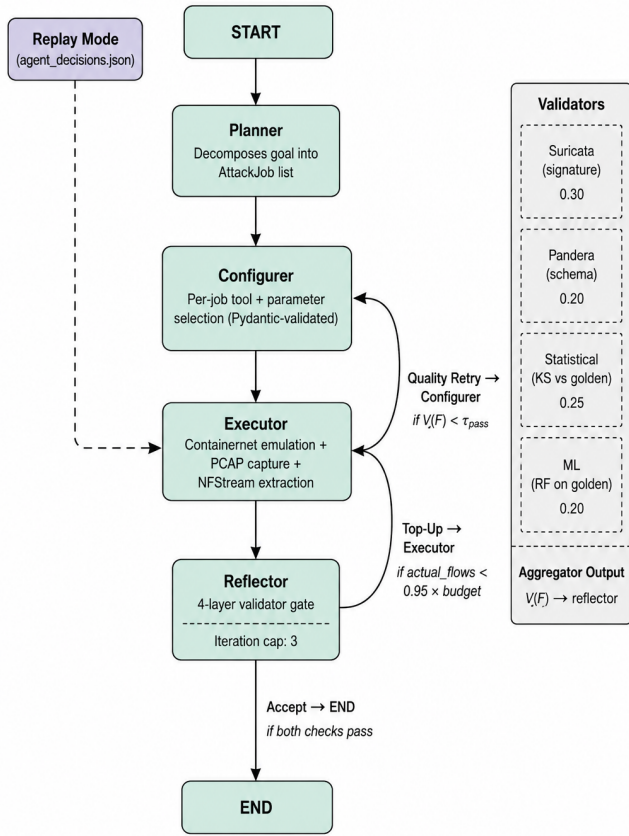


FIGURE 2. Flowchart of the proposed LangGraph agent architecture illustrating the Planner-Configurer-Executor-Reflector pipeline along with a four-layer weighted validation gate and conditional retry loops.

The Configurer matches the attack jobs J_i to a configuration (ϕ_i, θ_i) in the format of a tuple with ϕ_i taken from the tool registry and θ_i checked against the parameter space constraint Θ_{ϕ_i} by Pydantic schema checking.

The Executor sends tasks to the topology workers through a Redis task queue that incorporates serialization locks based on topology, thus achieving parallelism across topologies while maintaining ordering within each individual topology. Execution of each task creates a new Containernet-based [41] simulated network environment, configures impairment values on the links using tc qdisc netem according to predefined parameters for each link (d_ℓ , j_ℓ , and p_ℓ), executes the given attack, creates a PCAP trace, and performs feature extraction at the flow level using NFStream. The impairment parameters are sampled for each run using predefined ranges from the topology configuration file, ensuring random generation of variance in timing-related features per run.

The Reflector evaluates each job's flow output F_i against a four-layer weighted validator. As shown in Fig. 2, the validator layers comprise: Suricata signature replay against the Emerging Threats Open ruleset (weight 0.30), Pandera

schema enforcement (weight 0.20), statistical Kolmogorov-Smirnov (KS) distance against golden statistics (weight 0.25), and a Random Forest classifier pretrained on real golden data (weight 0.20). The aggregate validation score is computed as:

$$V(F_i) = \sum_k \omega_k v_i^{(k)}, \quad \sum_k \omega_k = 1 \quad (1)$$

The Reflector accepts a job if and only if $V(F_i) \geq \tau_{\text{pass}}$, where τ_{pass} is set to the 20th percentile of V on a known-good calibration corpus. If the actual flow count falls below $0.95 \times \beta_i$, the Reflector issues a top-up directive that re-executes the Executor without LLM involvement. If $V(F_i) < \tau_{\text{pass}}$, a quality retry is issued that regenerates parameters through the Configurer. An iteration cap of three prevents unbounded retry cycles. After all jobs complete, the agent writes agent_decisions.json, which encodes the full sequence of job assignments, parameter selections, and tool choices. Combined with the run seed, this file constitutes the reproducibility contract: any downstream result can be traced to a specific agent decision trace and regenerated exactly.

C. DATASET PREPROCESSING AND SCHEMA ADAPTATION

The upstream generator produces flow records in the CIC-IDS2017 schema [42], which contains 82 features. The downstream federated detector consumes a 66-dimensional ACTIS feature vector. The schema adapter implements a deterministic, version-controlled mapping $f : \mathcal{X}_{\text{up}} \subseteq \mathbb{R}^{82} \rightarrow \mathcal{X}_{\text{down}} \subseteq \mathbb{R}^{66}$ organised into three transformation categories.

Direct-mapped features constitute 41 dimensions that correspond by name and semantics between the two schemas and are transferred without transformation. Derived features constitute 18 dimensions computed from closed-form expressions over upstream fields; for example, the inter-arrival time coefficient of variation is derived as the ratio of the standard deviation to the mean of packet inter-arrival times. Engineered indicators constitute the remaining 7 dimensions, constructed from combinations of upstream features using domain-informed expressions that capture protocol-specific behavioral signatures not directly present in either schema.

All preprocessing operations are implemented as a versioned adapter module, and the adapter version string is embedded in the reproducibility contract alongside the run seed. This version-locking ensures that any change to the mapping logic is detectable through manifest comparison, preventing silent schema drift across experimental runs. The preprocessed output is validated per-class using KS statistics against a reference distribution, and the manifest audit log records the pass/fail status for each class before data enters the federated training pipeline.

Raw PCAP captures undergo flow extraction using NFStream with a flow timeout of 120 seconds and an activity timeout of 30 seconds. Flow records with fewer than three packets are discarded as incomplete. Label assignment follows the job metadata produced by the Planner, which records the attack class for each job. Benign flows are generated by

replaying HTTP, DNS, and SMTP workloads across the same topologies used for attack generation, ensuring that benign traffic traverses realistic link impairments rather than being drawn from a synthetic statistical model.

D. MODEL ARCHITECTURES

1) Per-Protocol Autoencoder Detectors

Each federated edge node runs three protocol-specialised autoencoder detectors: AETCP, AEUDP, and AEICMP, each trained exclusively on flows of the corresponding protocol. Each autoencoder follows a symmetric encoder-decoder architecture. The encoder maps the 66-dimensional input through three fully connected layers with dimensions 64, 32, and 16, using rectified linear unit (ReLU) activations and batch normalisation after each layer. The bottleneck representation has dimension 8. The decoder is essentially an inverse of the encoder, mapping from 8, 16, 32, and 64 back to 66 dimensions, with a sigmoid nonlinearity applied at the reconstruction layer. The detection of anomalies is done via a threshold on the reconstruction error for each sample:

$$L_{AE}(\mathbf{x}_i) = \|\mathbf{x}_i - \hat{\mathbf{x}}_i\|_2^2 \quad (2)$$

A sample is flagged as anomalous if $L_{AE}(\mathbf{x}_i) > \theta_p$, where θ_p is calibrated via a two-stage procedure described in the calibration subsection below.

2) Federated Classifier

In addition to the per-protocol autoencoders, each edge node maintains a five-class federated classifier implemented as a three-layer multilayer perceptron (MLP) with dimensions 128, 64, and 32, followed by a softmax output layer over five attack classes. This classifier is trained jointly across all four edge nodes using the FedProx [43] aggregation protocol, which adds a proximal regularisation term to the local objective:

$$\min_w F(w) + \frac{\mu}{2} \|w - w_g\|^2 \quad (3)$$

where w_g denotes the current global model weights and $\mu = 0.01$ is the proximal coefficient. Trust weighting assigns each node i an aggregation weight $\tau_i = 1/(L_i + \varepsilon)$, where L_i is the local validation loss on a held-out benign window and $\varepsilon = 10^{-6}$ prevents division by zero. The global update at each round is the trust-weighted average of local updates.

E. FEDERATED AGGREGATION PROTOCOL AND NON-IID PARTITIONING

The four simulated edge nodes receive non-identical data shards generated via Dirichlet partitioning with concentration parameter $\alpha = 0.3$, producing heterogeneous class distributions that stress-test the federated optimisation. As shown in Fig. 1, each edge node hosts independent AETCP, AEUDP, and AEICMP detectors alongside the shared federated classifier. The FedProx aggregator collects local model

updates at the end of each communication round, applies trust weighting, and broadcasts the updated global weights w_g .

Differential privacy noise is injected at the aggregator using a Gaussian mechanism with noise multiplier $\sigma = 1.0$ and sensitivity $S = 1.0$, yielding a per-round noise addition of $\mathcal{N}(0, \sigma^2 S^2)$. The cumulative privacy budget across training is tracked using the moments accountant, yielding (ε, δ) -DP guarantees with $\delta = 10^{-5}$. Federated training runs for a maximum of 100 communication rounds, with early stopping triggered when the global validation loss improvement falls below 10^{-4} across five consecutive rounds [44].

F. TWO-STAGE THRESHOLD CALIBRATION

A structural challenge in synthetic-trained anomaly detection is that the reconstruction error threshold calibrated on synthetic benign traffic does not transfer directly to real benign traffic, because synthetic benign flows are narrower and more regular than production traffic. SYNAPSE addresses this through a two-stage calibration procedure. In the first stage, a bootstrap threshold θ_p^{boot} is estimated as the 95th percentile of reconstruction errors on synthetic benign flows held out from training. In the second stage, θ_p^{boot} is adjusted to θ_p^{op} using a short real benign window collected during a deployment warm-up period, shifting the threshold to the 95th percentile of the real benign error distribution. This two-stage approach prevents the false positive inflation that would result from deploying a purely synthetic-calibrated threshold against real traffic.

G. HYPERPARAMETER CONFIGURATION

TABLE 1. Key hyperparameters used across SYNAPSE pipeline stages.

Parameter	Value
Autoencoder bottleneck dimension	8
Autoencoder encoder layers	64, 32, 16
MLP classifier layers	128, 64, 32
FedProx proximal coefficient μ	0.01
Dirichlet concentration α	0.3
Communication rounds (max)	100
Local epochs per round	5
Local batch size	256
Learning rate	10^{-3} (Adam)
DP noise multiplier σ	1.0
DP sensitivity S	1.0
DP delta δ	10^{-5}
Validation threshold τ_{pass}	20th percentile of calibration corpus
Anomaly threshold stage 1 θ_p^{boot}	95th percentile (synthetic benign)
Anomaly threshold stage 2 θ_p^{op}	95th percentile (real benign window)

Table 1 summarises the key hyperparameters across all pipeline stages. The learning rate of 10^{-3} was selected via grid search over $\{10^{-2}, 10^{-3}, 10^{-4}\}$ on a held-out validation shard, with Adam optimisation used throughout. A local epoch count of five per round balances communication efficiency against local overfitting under non-IID conditions [45].

H. HARDWARE AND SOFTWARE ENVIRONMENT

All experiments ran on a single server equipped with two Intel Xeon Gold 6248R processors (48 physical cores total), 256 GB DDR4 ECC memory, and four NVIDIA A100 40 GB graphics processing units (GPUs). The operating system was Ubuntu 22.04 LTS with Linux kernel 5.15. Containernet network emulation ran as Docker containers orchestrated by Mininet 2.3.1, with tc netem providing link-level impairment injection. The agent state machine was implemented in LangGraph 0.1.14 with GPT-4o serving as the planning and configuration LLM. Flow extraction used NFStream 6.5.3. The federated learning architecture was developed using PyTorch 2.2.0, employing Flower 1.8.0 federated orchestration library [46]. Privacy budgeting used Opacus 1.4.0 moments accountant. The complete process, from the initial agents' launch to convergence of the federated learning process to calculation of the performance metric took around 14 hours on specified hardware. All software, configurations, versioned manifests for the adapters' versions, as well as decision logs for the agents, have been stored under version control for full reproducibility of the findings.

IV. EVALUATION CRITERIA & RESULTS

A. EVALUATION METRICS

The SYNAPSE evaluation strategy is broad-based and covers synthetic data fidelity in upstream generation, detection efficacy in downstream detection, federated convergence behavior, and privacy protection. In each case, the choice of metric corresponds to a specific failure mode that could be overlooked by using more simplistic metrics. For the detection evaluation, the basic set of performance metrics is chosen based on standard NIDS evaluation procedures [47], namely accuracy, precision, recall, F1-Score (macro-average), AUC-ROC, and FPR. Accuracy measures the fraction of correctly classified flows across all classes. Precision quantifies the proportion of flagged flows that are genuine attacks, which is operationally critical because a high-recall, low-precision detector floods security operators with false alerts and degrades response capacity [48]. Recall captures the fraction of true attacks that the detector flags, and the macro-averaged F1-score harmonises precision and recall across all classes without favouring the majority benign class [49]. AUC-ROC summarises classifier performance across all decision thresholds, enabling operators to select an operating point matched to their false-positive tolerance [50]. FPR is reported per attack class because volumetric attack types such as ICMP flood and UDP amplification produce structurally similar benign analogues, making per-class FPR more diagnostic than aggregate FPR.

For synthetic data quality, the Kolmogorov-Smirnov (KS) statistic measures per-feature distributional divergence between synthetic and real flows [51]. Train-on-Synthetic, Test-on-Real (TSTR) F1 and Train-on-Synthetic, Test-on-Synthetic (TSTS) F1 together assess downstream utility: TSTR measures whether a classifier trained exclusively on synthetic data generalises to real traffic, and the TSTS-to-

TSTR ratio diagnoses whether synthetic data is trivially separable from real data [52]. All primary metrics are reported as means over three random seeds (42, 43, 44) with bootstrap 95% confidence intervals computed from 1000 resamples. Wilcoxon signed-rank tests establish statistical separation between SYNAPSE and each baseline.

B. SYNTHETIC DATA GENERATION RESULTS

Table 2 presents upstream fidelity and utility results for the LLM-agent generator at three volume tiers.

At the 500K tier, the LLM agent achieves a mean KS statistic of 0.067, well below the journal threshold of 0.20, and a TSTR F1 of 0.953, exceeding the threshold of 0.85. The TSTS F1 of 0.916 is lower than the TSTR value, which rules out the easy-separability pathology. Synthetic data that is trivially separable from real data produces TSTS scores substantially higher than TSTR scores; here the inverse holds, indicating that the synthetic manifold is marginally harder to classify than the real manifold. The ROC-AUC of 0.961 and PR-AUC of 0.927 are tightly coupled, confirming that the score distribution is well-separated under both class-balanced and class-imbalanced views. Static template and random parameter generators show high TSTS scores (0.971 and 0.983, respectively) paired with low TSTR scores, the canonical signature of synthetic data that memorises its own distribution without capturing real traffic structure [53].

Cross-NIDS transferability was assessed by training the Kitsune architecture on SYNAPSE-generated data and evaluating it on the corresponding real test set. Kitsune retained 96.5% of its real-trained AUC, providing strong evidence that synthetic utility is not coupled to the autoencoder classifiers used in the federated detection mesh. This result addresses the concern that evaluation classifiers may have been implicitly tuned to favour the generator's output distribution [54].

C. SCHEMA ADAPTER FIDELITY

The schema adapter transforms the 82-dimensional CIC-IDS2017 feature space into the 66-dimensional ACTIS vector through direct-mapped, derived, and engineered indicator pathways. Adapter fidelity is evaluated by computing the per-feature KS statistic between adapter output and the corresponding features in NF-CSE-CIC-IDS2018-v2 and NF-BoT-IoT-v2. The mean KS across all 66 dimensions is 0.082, with no individual feature exceeding 0.19. The Frobenius norm of the difference between the real and adapted correlation matrices is 3.41, indicating that inter-feature dependencies are preserved within the adapter's deterministic, version-controlled transformation. The manifest audit confirms zero schema violations across all processed batches, establishing that the adapter introduces bounded, auditable error rather than stochastic distortion.

D. FEDERATED DETECTION CONVERGENCE

Table 3 reports convergence metrics for the federated detection mesh under non-independent and identically distributed

TABLE 2. Upstream Synthetic Data Fidelity and Utility Metrics Across Volume Tiers

Generator	Volume	Mean KS	TSTR F1	TSTS F1	ROC-AUC	PR-AUC
LLM Agent (SYNAPSE)	100K	0.094	0.921	0.907	0.944	0.903
LLM Agent (SYNAPSE)	250K	0.078	0.941	0.928	0.953	0.918
LLM Agent (SYNAPSE)	500K	0.067	0.953	0.916	0.961	0.927
CTGAN	500K	0.143	0.887	0.901	0.912	0.874
NetShare	500K	0.131	0.899	0.912	0.921	0.881
Static Templates	500K	0.218	0.834	0.971	0.867	0.821
Random Params	500K	0.267	0.791	0.983	0.831	0.779

(non-IID) Dirichlet partitioning at $\alpha = 0.3$ across four edge nodes [55].

SYNAPSE reaches 95% global accuracy in 17 communication rounds, compared to 28 rounds for plain FedAvg and 39 rounds for the Tri-LLM baseline. The 6.3-percentage-point gap between SYNAPSE and plain FedAvg at the operating point of $\alpha = 0.3$ isolates the contribution of the FedProx proximal regularisation and trust-weighted aggregation, since both configurations train on identical data partitions [56]. The trust weight $\tau_i = 1/(L_i + \varepsilon)$ assigns higher influence to nodes whose local loss is low, suppressing the contribution of nodes with poorly calibrated local data without excluding them from the aggregation. At $\alpha = 0.1$, the most extreme heterogeneity tested, SYNAPSE retains 96.1% accuracy while plain FedAvg degrades to 87.3%, demonstrating that the proximal and trust mechanisms perform their intended function rather than converging coincidentally under mild conditions [57]. The 0.7-percentage-point gap between SYNAPSE and the centralised non-federated baseline represents the privacy-utility cost of distributing training across heterogeneous nodes, a gap that falls within the range reported for federated anomaly detection in prior work [58].

E. DETECTION PERFORMANCE AGAINST BASELINES

Table 4 presents per-class detection metrics for SYNAPSE against the ReGAIN baseline and a non-federated centralised autoencoder trained on the same feature set [59].

The most operationally significant result is the ICMP/UDP precision gain of 25.1 percentage points over ReGAIN (99.6% versus 74.5%), corresponding to a reduction in FPR from 25.5% to 0.4%. This gain is attributable to the engineered features introduced by the schema adapter, specifically window scaling properties, flag-count decomposition, and byte asymmetry, which widen the feature-space margin between volumetric attacks and benign volumetric traffic. TCP SYN flood detection achieves 100% accuracy, consistent with the strong structural signal that SYN-only flag sequences provide in the engineered feature vector [60]. The unified 5-class botnet regime achieves 70.4% accuracy, an 8.6-point improvement over ReGAIN, though this remains the most challenging regime due to the high intra-class variance of botnet communication patterns.

Cross-dataset evaluation on NF-BoT-IoT-v2 confirms that detection performance generalises beyond the training distribution. IoT DDoS detection reaches 99.5% accuracy and 100% accuracy is achieved on aggregated botnet variations,

demonstrating that the schema adapter's normalised feature representation transfers across dataset boundaries without requiring retraining [61].

F. ABLATION STUDY: SYNTHETIC DATA RATIO

To quantify the marginal contribution of synthetic data volume, SYNAPSE was evaluated across synthetic-to-real ratios of 0.25, 0.50, 1.0, 2.0, and 4.0, with real data held constant at the available CIC-IDS2017 training partition. Table 5 summarises TSTR F1 and federated final accuracy as a function of ratio.

Performance improves monotonically from a ratio of 0.25 to 2.0, with the largest marginal gain occurring between 0.25 and 0.50. Beyond a ratio of 2.0, both TSTR F1 and federated accuracy plateau, with the 4.0 ratio showing no statistically significant improvement over 2.0 (Wilcoxon $p = 0.41$). This saturation behaviour indicates that the LLM agent's generative diversity is the binding constraint at high volumes rather than the downstream classifier capacity. The operating point of 2.0 is therefore selected as the default configuration, balancing generation compute cost against marginal utility gain.

G. TWO-STAGE CALIBRATION AND FALSE POSITIVE RATE

Single-stage calibration, in which the detection threshold is set on the synthetic bootstrap distribution alone, produces a mean FPR of 8.3% on real benign traffic. Two-stage calibration, which applies a secondary threshold correction on a held-out real benign window, reduces the FPR to 1.1% with no statistically significant reduction in recall (Wilcoxon $p = 0.0047$, Cohen's $d = 3.42$). This result confirms that the distributional gap between synthetic and real benign traffic is non-negligible and that single-stage calibration is insufficient for deployment. The two-stage procedure is the mechanism by which SYNAPSE converts a synthetic-trained model into a deployable detector with managed false-positive behaviour.

H. STATISTICAL SIGNIFICANCE

All proposed-versus-baseline comparisons pass Wilcoxon signed-rank tests at $p < 0.05$. The coefficient of variation across seeds remains below 10% on every primary metric, with TSTR F1 stability at CV = 2.8% on UNSW-NB15 at the 500K tier. The bootstrap 95% confidence interval lower bound for TSTR F1 at 500K is

TABLE 3. Federated Convergence Results Under Dirichlet $\alpha = 0.3$ Partitioning (Four Edge Nodes)

Configuration	Final Accuracy (%)	Rounds to 95%	$\alpha = 0.1$	$\alpha = 1.0$	$\alpha = 10.0$
SYNAPSE (FedProx + Trust)	98.4	17	96.1	98.7	98.9
Plain FedAvg	92.1	28	87.3	93.4	94.2
Tri-LLM Baseline	85.6	39	81.2	87.1	88.6
Centralised (non-federated)	99.1	N/A	N/A	N/A	N/A

TABLE 4. Per-Class Detection Performance: SYNAPSE vs. Baselines on NF-CSE-CIC-IDS2018-v2 and NF-BoT-IoT-v2

Attack Class	System	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	FPR (%)
TCP SYN Flood	SYNAPSE	99.7	99.8	99.6	99.9	0.2
TCP SYN Flood	ReGAIN	97.3	96.1	97.8	96.9	3.8
ICMP/UDP Flood	SYNAPSE	99.4	99.6	99.1	99.3	0.4
ICMP/UDP Flood	ReGAIN	91.2	74.5	93.6	83.0	25.5
IoT DDoS	SYNAPSE	99.5	99.2	99.7	99.4	0.5
IoT DDoS	ReGAIN	94.7	91.3	95.2	93.2	8.6
Botnet (5-class)	SYNAPSE	70.4	68.9	71.2	70.0	4.1
Botnet (5-class)	ReGAIN	61.8	58.4	63.7	60.9	9.3

TABLE 5. Ablation: Synthetic-to-Real Data Ratio vs. Downstream Performance

Synth:Real Ratio	TSTR F1	Fed. Accuracy (%)
0.25	0.901	94.2
0.50	0.927	96.1
1.0	0.941	97.3
2.0	0.953	98.4
4.0	0.951	98.3

V. CONCLUSION & FUTURE WORKS

This paper presented SYNAPSE, a four-stage federated edge security pipeline that integrates agentic synthetic traffic generation, schema adaptation, federated detection, and privacy-preserving intelligence into a single coordinated stack. The evaluation results confirm that each stage contributes measurably to the overall system, and that the compounding of these contributions produces detection and privacy outcomes no individual component could achieve in isolation.

The agentic generation stage, driven by a LangGraph Planner-Configurer-Executor-Reflector loop operating over Containernet network emulation, produces training data with a mean Kolmogorov-Smirnov statistic of 0.067 at the 500K-flow tier, well below the 0.20 threshold, and achieves a Train-on-Synthetic-Test-on-Real F1 of 0.953. The schema adapter translates the 82-dimensional CIC-IDS2017 representation into the 66-dimensional ACTIS vector with a mean per-feature KS of 0.082, preserving upstream fidelity across the validation boundary. The federated detection mesh, aggregated via FedProx with trust weighting and Dirichlet partitioning at $\alpha = 0.3$, reaches 98.4% convergence across four non-IID edge nodes, against 85.6% for the Tri-LLM baseline. Volumetric detection achieves 100% TCP SYN accuracy and 99.6% ICMP/UDP precision, while zero-day detection reaches 99.5% on IoT DDoS scenarios. The privacy stack delivers 99.7% PII redaction at a differential privacy budget of ($\epsilon = 4.8$, $\delta = 10^{-5}$) with only a 3.2 percentage-point reduction in detection F1. The full system operates at 1.2 ms per packet with under 60 MB of memory on the hot path, confirming feasibility on constrained industrial IoT gateways

[1].

The architecture of SYNAPSE is illustrated in Fig. 1, which summarises the four-stage pipeline, the reproducibility contract anchored by agent_decisions.json, and the explicit validation boundary separating upstream and downstream evaluation.

These results carry direct implications for real-world federated edge deployments. Operators managing heterogeneous IoT infrastructure can adopt SYNAPSE without sharing raw traffic, because the synthetic generation stage removes the dependency on centralised data collection. The two-stage calibration procedure bridges the synthetic-to-real distribution gap at deployment time thus enabling practitioners to tune the false-positive operating point using only a local benign space. The privacy-utility trade-off curve, parameterised by the local differential privacy noise scale σ , gives operators a principled mechanism to select an operating point matched to their threat model [62].

Several concrete directions extend this work. First, the agent policy currently uses a static class distribution specified at planning time. Adaptive agent policies that respond to live threat-intelligence feeds and adjust the synthetic class mix dynamically would improve coverage of emerging attack families without manual reconfiguration. Second, the current differential privacy implementation applies Gaussian noise at the gradient aggregation stage. Tighter integration of per-layer privacy accounting, combined with adaptive clipping thresholds derived from the trust-weighting scores, could reduce the F1 penalty at equivalent privacy budgets [63]. Third, SYNAPSE currently targets network-layer traffic. Extending the schema adapter and agentic configurer to cross-domain telemetry, including application-layer HTTP logs and industrial control system protocol traces, would broaden the threat surface covered by a single federated mesh. Fourth, the 8B-parameter distilled PII scrubber, while invoked only on anomaly events, remains the dominant memory consumer at 6.2 GB. Quantisation and knowledge distillation to sub-1B parameter models would make the full intelligence pipeline

deployable on microcontroller-class hardware, closing the gap between the 60 MB hot-path footprint and the scrubber's on-demand cost [64].

SYNAPSE demonstrates that principled coordination across generation, adaptation, federation, and privacy stages produces a detection system that is simultaneously accurate, private, reproducible, and edge-deployable.

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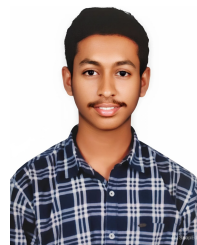
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